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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

First / Second Semester B.E. (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21PH102 / 20PH102 - ENGINEERING PHYSICS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

List of constants: Velocity of light, $c=3 \times 10^8 \text{ ms}^{-1}$, Planck's constant, $h=6.63 \times 10^{-34} \text{ Js}$,
Electron mass, $m=9.11 \times 10^{-31} \text{ kg}$, Electron charge, $e=1.602 \times 10^{-19} \text{ C}$,
Permittivity of vacuum, $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$, Boltzmann constant, $k=1.38 \times 10^{-23} \text{ J/K}$.
Avogadro number, $N_A = 6.023 \times 10^{26} \text{ kg mole}^{-1}$.

Unit – I		Marks	BT*	CO*	PO*
1. a)	What are Miller indices? Mention their salient features.	6	L*2	1	1, 2
b)	Derive Schrodinger's time independent one-dimensional wave equation for a particle of mass m with energy E .	10	L2	2	1, 2
c)	The lattice constant for a unit cell of aluminum is $4.031 \text{ \AA}$. Calculate the interplanar distance of $(2 \ 1 \ 1)$ plane.	4	L3	2	1, 2
2. a)	What are matter waves? Mention its characteristics.	4	L2	2	1, 2
b)	What is inter planar distance? Obtain an expression in terms of lattice parameter and miller indices for a cubic crystal.	6	L2	1	1, 2
c)	What are X-rays? Explain the origin of continuous X- ray spectrum.	6	L2	2	1, 2
c)	An X-ray machine has an accelerating potential of 25 kV . Find the shortest wavelength present in the X-ray spectrum.	4	L3	2	1, 2
3. a)	Describe the crystal structure of ZnS .	6	L2	2	1, 2
b)	Solve the Schrodinger wave equation for a particle in one dimension potential well of infinite height and discuss about energy Eigen values and Eigen functions.	10	L2	1	1, 2
c)	Calculate the de Broglie wavelength associated with an electron subjected to a potential difference of 1.25 kV .	4	L3	1	1, 2
Unit – II					
4. a)	Explain the principle and working of Zener diode.	6	L2	3	1, 2
b)	What is Hall Effect? Explain how Hall field is generated in an n-type semiconductor and obtain an expression for the Hall voltage, Hall coefficient and mobility of charge carriers.	10	L2	3	1, 2
c)	The Hall coefficient of a specimen of a doped silicon is found to be $3.66 \times 10^{-4} \text{ m}^3/\text{C}$. The resistivity of the specimen is $8.93 \times 10^{-3} \Omega \text{ m}$. Find the mobility and density of the charge carriers.	4	L3	3	1, 2
5. a)	What are superconductors? Explain the effect of magnetic field in superconductors.	6	L2	4	1, 2
b)	Explain the effect of temperature on the Fermi level in an n-type and p-type semiconductors.	10	L2	3	1, 2
c)	The critical temperature and critical field for superconducting lead are 7.2 K and 800 gauss respectively. What will be the temperature upto which lead will be in superconducting state in a magnetic field of 400 gauss ?	4	L3	4	1, 2

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| 6. | a) | Derive an expression for conductivity of intrinsic semiconductor. | 6 | L2 | 4 | 1, 2 |
| | b) | Explain the types of superconductors. Mention various applications of superconductors. | 10 | L2 | 4 | 1, 2 |
| | c) | Calculate the resistivity of intrinsic germanium at 300K if the intrinsic carrier density is $2.5 \times 10^{19} \text{ m}^{-3}$ assuming electron and hole mobilities of $0.38 \text{ m}^2\text{V}^{-1}\text{S}^{-1}$ and $0.18 \text{ m}^2\text{V}^{-1}\text{S}^{-1}$ respectively. | 4 | L3 | 3 | 1, 2 |

Unit – III

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| 7. | a) | What is a LASER? Explain the stimulated emission of radiation in laser. | 6 | L2 | 5 | 1, 2 |
| | b) | Describe the construction and working of a Ruby laser. Mention its applications. | 10 | L2 | 5 | 1, 2 |
| | c) | Calculate the wavelength of emission from a GaAs semiconductor laser whose band gap energy is 1.44 eV. | 4 | L3 | 5 | 1, 2 |
| 8. | a) | What is an optical fiber? Explain the principle on the basis which optical transmission is achieved through a fiber? | 6 | L2 | 5 | 1, 2 |
| | b) | What are the different modes of propagation possible in optical fibers? Explain in detail with necessary diagrams. | 10 | L2 | 5 | 1, 2 |
| | c) | The refractive index of core and cladding are 1.50 and 1.48 respectively in an optical fiber. Find the numerical aperture and angle of acceptance. | 4 | L3 | 5 | 1, 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
